

Soft Robotic Finger with compliant locking mechanism

Currently, there are many underactuated robotic hands which have been developed that can perform a wide range of functional grips and whole hand motions. Similarly, rigid robots often integrate internal mechanisms that can make a lot of motions like abduction and adduction. However, there is still a gap in the field, soft robotic hands that can achieve a full range of natural finger motions including flexion, extension, abduction and adduction remain limited.

To address this challenge, this thesis proposes the design of a soft robotic finger with integrated compliant locking mechanisms. Compliant mechanisms are mechanisms which uses elastic deformation of materials rather than traditional rigid joints. These mechanisms will not have the problems which are being faced by rigid mechanisms like friction and loss of energy. which are faced by the regular mechanisms. These mechanisms can be embedded at different parts of the finger to have an adaptable locking mechanism.

These mechanisms can be made in the base for the fingers to make the motion of abduction and adduction. It can be implemented along the finger body on the on each of the phalanges to make the motion of flexion and extension with selective motion constrains.

I would also like to test out different materials like PLA, Nylon (If available) and TPU for the compliant mechanisms and for the fingers I would like to try using TPU and Silicone. Using these materials, I would like to see the perfect combination of materials could make this finger work.

After the finger design is done through the fabrication stage, I would like to try to actuate it using motors, test it and control it. I think we would need 4 motors to actuate the finger: 3 motors for each of the phalanges to deploy and anti-deploy the compliant locking mechanism, and 1 servo motor for the flexion, abduction and adduction.

The servo motor must be connected to 2 tendons and is must also have a gear mechanism with 2 modes one where the both tendons are being pulled or pushed (C-C/A-A) and then the second mode when the motor is pulling one string and pushing the other(C-A/A-C).

If the time permits, I would like to continue the project even forward by creation a palm and 5 fingers to make a soft robotic hand with a wrist.

